

Test strategy (module 6)

PaxAuto

1) Feature – 4: Users can stay safe with following traffic light and collision detection while traveling on our shuttle.

2) Feature Owner: Sahil Rajesh Patil

3) Feature Description - (interface to user)

- Shuttle rides normally to designated location.
- While on ride if traffic light is RED, shuttle stops before traffic signal.
- While on ride if traffic light is GREEN, shuttle proceeds to its destination.
- While at intersection if shuttle finds traffic lights disabled shuttle will stop before intersection.
- If an obstacle appears in path, on straight path or curve path, shuttle will stop in front of obstacles.
- User experiences the safe behaviour of shuttle as it follows traffic rules and avoid collision with obstacles in path, maintaining safe behaviour for user journey.

a) KPIs:

KPI-1 – Shuttle identify the traffic signal state and stop for red signal (state – 2,3)

KPI-2 – Shuttle identify the traffic signal state red and green (state – 1) red (state – 2,3) green

KPI-3 – Shuttle should subscribe all obstacle around it and then filter one which are in shuttle current journey path and send Boolean to lateral and longitudinal control to stop before collision.

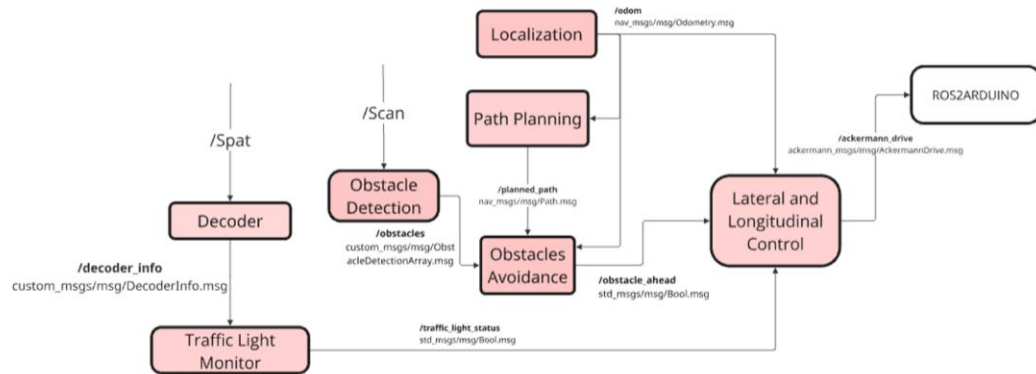
4) Functionality

1. F4.1 Traffic Light Monitor component
 - Subscribing SPaT(M) message from Ad_Infrastructure – intersect that message to decode light state (1 to 9) → traffic light monitor identifies state and publish int to lateral and longitudinal control to know light state (RED/ GREEN)
2. F4.2 Collision Avoidance component
 - Subscribe custom messages from obstacle detection knowing obstacles distance, coordinate and id which later filters by obstacle avoidance which identify obstacles in path and publish Boolean to lateral and longitudinal control to stop.

5) Architecture:

- Link to system architecture - ([link](#))
- **Component involved with GIT link:** lidar, [obstacle detection](#), [obstacle avoidance](#), ad infrastructure, [decoder](#), [traffic light monitor](#), [path planner](#), [localization](#).
 - Obstacle detection: **input** - /scan, **output** - / obstacles
 - obstacle avoidance: **input** - /odom , /planned_path , / obstacles, **output** - / obstacle _ahead
 - decoder: **input** - /spat, **output** - /decoder_info
 - traffic light monitor: **input** - /decoder_info, **output** - /traffic_light_status

Component Responsible



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6) Interfaces to other features:

Other features can be accessed via provided link ([miro link](#))

Other features	Description	Link to feature F4
The user can experience a seamless and smooth driving journey between requested locations.	The shuttle generates path using path planner, from its current position to requested pickup or drop-off location in form of path coordinates (x,y) which is later used by lateral and longitudinal control to follow those coordinates one by one.	The path coordinates are used by obstacle avoidance component to verify if obstacles coordinates are in specific detection range. The traffic light monitors after knowing traffic signal state it publishes int to lateral and longitudinal control to stop or proceed.

7) Test strategy

a) Test basis

Artifacts:

- All dependent features must be working
- Required subscriptions across components: /spat, /obstacles, /odom, /path_planner, /decoder_info, /scan. All this must be publishing with data
- Path planner and lateral + longitudinal control provide motion commands to the shuttle
- AD infrastructure always provides spatem message, which can be decoded by the decoder
- /Obstacles provides left and right boundary points (x, y) and they must be perfectly aligned in the map frame
- LiDAR must be available, properly aligned, and correct roll configured in Model City

b) Test techniques

The following black box technique are used.

- Scenario based testing

c) Scenario for testing considering KPIs

1. KPI-1 – Shuttle identify the traffic signal state and stop for red signal (state – 2,3) (5 x RUNs)

Metrics – 100% if shuttle completes pass criteria

method – test run of shuttle on red signal at intersection id (x, y).

pass criteria –

- shuttle stops on red signal state every time
- Shuttle stops before intersection not after intersection

Results –

- Shuttle stopped every time signal turned red before intersection in every run

2. KPI-2 – Shuttle identify the traffic signal state red and green - (state –2,3) red, (state – 1) green (5 x RUNs)

Metrics –100% if shuttle completes pass criteria

Method - test run of shuttle stops on red and move on green signal at intersection id (x, y)

Pass criteria –

- KPI-1 should be completed successfully
- After signal turn to green shuttle should resume journey from where it left.

Results –

- Shuttle stopped at red and continued at green every time continuing path from where it left

3. KPI-3 - Shuttle should subscribe all /obstacle, around it and then filter one which are in shuttle current journey path and send int to lateral and longitudinal control to stop before collision. (5 x RUNs)

Metrics –100% if shuttle completes pass criteria

Method – shuttle will run on curve path and straight path with obstacle on path

Pass criteria -

- Shuttle should process all obstacles position in correct frame to correctly identify distance and coordinates
- Shuttle should make a detection range of 40 degree from front axle and bubble around path coordinates points to verify if obstacle is in range or not
- Shuttle should stop whenever it detects obstacle in range before 0.50m to avoid collision

Result –

- Shuttle started on curve path faced plant and stopped on curve next it stopped on curve seeing mannequin and on straight path it stopped before box in 4 consecutive runs.
- Shuttle collided with obstacle in turn as while making turn obstacle came too close, below detection range of lidar ($s < 0.5m$) which cause collision.

Only above-mentioned criteria are required to fulfil feature any tweak in criteria or any other situation not triggering same criteria will cause feature function failure entirely or partially.

d) Test case

- Verify stop and go function at traffic signal intersection based on signal state
- Verify stopping in front of Obstacles when detected in path
- Measure the success rate across repeated scenario execution
- More information is documented in test specification document

e) Coverage

Scenario coverage

- Metric: 100%
- Reason: all defined operation scenarios are executed meeting all pass criteria

KPI coverage

- Metrics: 90%
- Reason: Each KPI is directly evaluated by at least one scenario-based system test case

f) Test environments

Model city with functioning Opti track and lidar onboard on shuttle.